ARAVIND KUMAR R

Bengaluru, Karnataka, India | aravindkumar.ai@outlook.com | LinkedIn | GitHub | aravindkumar.ai

TECHNICAL SKILLS

AI Agents & Generative AI:	Autonomous AI Agents, Claude Agent SDK, Claude Code, Multi-Agent Systems, AI Red Teaming & Pentesting, Tool Calling / Function Calling, LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning (PEFT - LoRA, QLoRA) Small Language Models (SLM), Large Language Models (LLM)
Deep Learning & Machine Learning:	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, EmbeddingGemma, Hierarchical Multi-label Classification, Quantization (GGUF, AWQ, INT8/FP16), Vector Embeddings
Natural Language Processing (NLP):	BERTopic, KeyBERT, BERT, BART, T5, LayoutLM, SpaCy, NLTK, Speech Recognition (ASR / STT), Text-to-Speech (TTS), Multi-lingual AI Dubbing, RASA, DialogFlow
Computer Vision:	OpenCV, YOLO (v5/v8), NVIDIA DeepStream, Object Detection, Instance Segmentation, Real-time Video Analytics, Image Classification
MLOps & High-Performance Serving:	vLLM, NVIDIA TensorRT, Triton Inference Server, Docker, FastAPI, Flask, Linux, CI/CD, Streamlit, GPU Acceleration
Cloud & Distributed Data:	AWS (SageMaker, Lambda, EC2, S3, CloudWatch), Google Cloud Platform (GCP), Neo4j (Graph DB), Vector Databases (Pinecone, Chroma, Milvus), PostgreSQL, MySQL, MongoDB, Elasticsearch
Programming Languages:	Python, C++, SQL, Bash

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer - KnowBe4 **Feb 2024 - Present**

Bengaluru, Karnataka, India

- **AIDA (Artificial Intelligence Defense Agents):** Architected and deployed agentic capabilities within AIDA, building automated phishing template generation for adaptive phishing simulations and personalized training recommendations driven by employee **RiskScore** metrics.
- **Agent Security & Prompt Injection Defense:** Engineered a prompt injection classifier based on **EmbeddingGemma** with custom classification heads, optimized inference pipelines using **vLLM** and GPU acceleration, boosting inference throughput by **6x** with low latency.
- **LangChain LLM Workflows:** Designed and deployed LangChain-based production pipelines for automated quiz generation from compliance policy documents and course modules, and built a multi-level **hierarchical multi-label, multi-class classification engine (124 classes)** for training module categorization and content discovery.
- **Agentic Workflows & Safety:** Spearheaded agentic evaluation pipelines and developer automation utilizing the **Claude Agent SDK** and **Claude Code** for automated agent auditing and internal security testing.

Machine Learning Engineer - LearnTube.ai (CareerNinja (BeNinja Solutions)) **Oct 2021 - Aug 2023**

Mumbai, Maharashtra, India

- Spearheaded core AI/ML systems powering an AI-driven learning platform serving over **100K learners**, reducing end-to-end curriculum and content generation cycles from days to under 2 hours.
- Engineered, deployed, and scaled a **Neo4j Knowledge Graph** based content recommendation system connecting skills, learning objectives, and course videos to deliver dynamic personalized learning pathways.
- Built automated AI video dubbing pipeline orchestrating Whisper speech-to-text (ASR), neural machine translation, and multilingual voice synthesis to localize educational video content into regional Indian languages.
- Architected an NLP discovery engine leveraging **BERTopic, KeyBERT, and automated web crawlers** to ingest, topic-model, and rank educational YouTube videos with sub-second retrieval latency.
- Developed production Generative AI features including Instant Doubt Solving, AI Question Generation, and Summarization using GPT-4, Claude, LLaMA, and open-source foundation models.

Senior Software Development Engineer - Applied AI - DIATOZ Solutions **May 2019 - Oct 2021**

Bengaluru, Karnataka, India

- Led an engineering team of 4 in designing, deploying, and maintaining AI engines for **E2Ehiring** (e2ehiring.com), an AI recruitment platform featuring automated resume information extraction (LayoutLM / OCR), candidate-job matching algorithms, and automated interview assessment scoring.
- Delivered an **8x inference speedup** across real-time video analytics pipelines by migrating deep learning models to **NVIDIA DeepStream, TensorRT, and Triton Inference Server** with FP16/INT8 model quantization.
- Led a 3-engineer squad to build and maintain **Ezevision**, a SaaS traffic analytics platform providing real-time multi-camera vehicle detection, speed estimation, and congestion monitoring.
- Built multi-lingual customer support conversational agents supporting 4 Indian languages for a tier-1 Indian infrastructure enterprise utilizing RASA, DialogFlow, and custom BERT intent classifiers.
- Authored production deployment standards and mentored engineers on containerized microservice architectures using Docker, FastAPI, and GPU acceleration.

CERTIFICATIONS

Claude Certified Architect - Foundations (CCAF) - Aug 2026 • **Extensive and Re-imagined AI Program v3 - AI Course** - The School of AI (2024 - 2025) • **AWS Certified AI Practitioner** - Amazon Web Services (Active / July 2026) • **AWS Certified Cloud Practitioner** - Amazon Web Services (Active / June 2025)

HONORS & AWARDS

Winner - KnowBe4 AI Hackathon Bengaluru (2026): Built an autonomous Red Team Hacking Agent evaluating AI agent security, prompt injection defenses, and vulnerability assessment. • **Runners Up - Engineering Innovation Summit (2024):** Recognized for excellence in architecting and deploying real-world generative AI and agentic systems. • **Best Employee Award (2022) - LearnTube.ai / CareerNinja:** Conferred for outstanding impact in scaling GenAI features and Neo4j graph recommendations to 100K+ users.

PUBLICATIONS

AURATER: Autonomous Unmanned Rover with Advanced Trash Eliminating Robot *International Conference on Recent Advances in Engineering and Technology (ICRAET - IEEE Sponsored)*

EDUCATION

Bachelor of Engineering (B.E.) in Mechatronics Engineering, Sri Krishna College of Engineering and Technology, Coimbatore, Tamil Nadu, India • CGPA: 8.37 / 10 (NIRF Top 100 Engineering Institution) 2015 - 2018

Diploma in Mechanical Engineering, Nachimuthu Polytechnic College, Coimbatore, Tamil Nadu, India • Cumulative Percentage: 88% 2012 - 2015